Lab: Agent Profile -- What Makes You Tick?

Objective

> **Learning Goal:** Analyze yourself and others as agents -- things that sense the world, build mental models, and take actions to achieve goals.

This is a **group challenge** designed for middle school students (Grades 6-8).

Materials Needed

- Index cards or paper
- Colored pens or pencils
- A partner or small group
- Lab journal or notebook

Part 1: Agent vs. Object (10 min)

Not everything is an agent. A rock just sits there. But YOU make decisions, react to what happens, and try to reach goals. That makes you an agent.

Sort Challenge: Decide whether each item below is an agent or just an object. Write A (agent) or O (object) and explain why.

Item	Agent or Object?	Why?
A thermostat		
A basketball		
Your pet (or a pet you know)		
A vending machine		
A plant growing toward light		
Your phone's autocorrect		
A puddle of water		
A player in a video game (NPC)		

Write your response here

Part 2: Build Your Agent Profile (15 min)

Every agent has three core abilities: **sensing** (taking in information), **modeling** (building expectations about the world), and **acting** (doing things to achieve goals).

Create an "Agent Profile Card" for yourself:

AGENT PROFILE: ___ (your name)

Feature	Your Details
Sensors (How do you gather information?)	
Top 3 Goals Right Now	
Internal Model (What do you expect to happen today?)	
Favorite Actions (What do you do most to reach your goals?)	
Biggest Source of Surprise This Week	
How You Handle Surprise	

Write your response here

Part 3: Agent Comparison (10 min)

Swap profile cards with a partner. Compare your two agent profiles.

Answer these questions together:

1. What goals do you have in common?
2. Where do your internal models differ? (Do you expect different things from the same situations?)
3. When you both face the same surprise, do you react the same way or differently?

Write your response here

Part 4: Design a Simple Agent (10 min)

Imagine you are designing a robot agent for your school. It needs to navigate the hallway between classes.

Design Element	Your Robot Agent
What sensors does it need?	
What is its goal?	
What predictions should it make?	
What actions can it take?	
What would surprise it?	
How should it handle surprise?	

Write your response here

Reflection Table

Question	Your Answer
What makes something an agent vs. just an object?	
What are the three core abilities of any agent?	
How is a thermostat like you? How is it different?	
Why do different agents react differently to the same situation?	
How does Active Inference describe what agents do?	

Write your response here